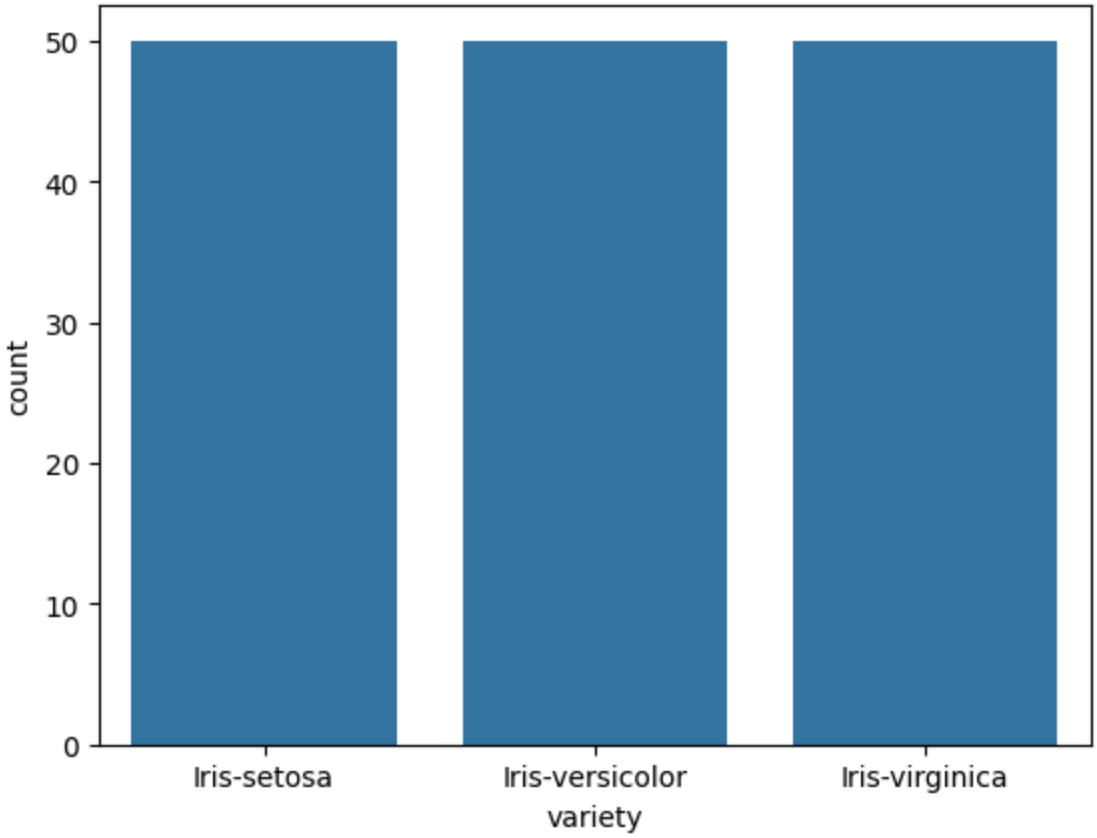
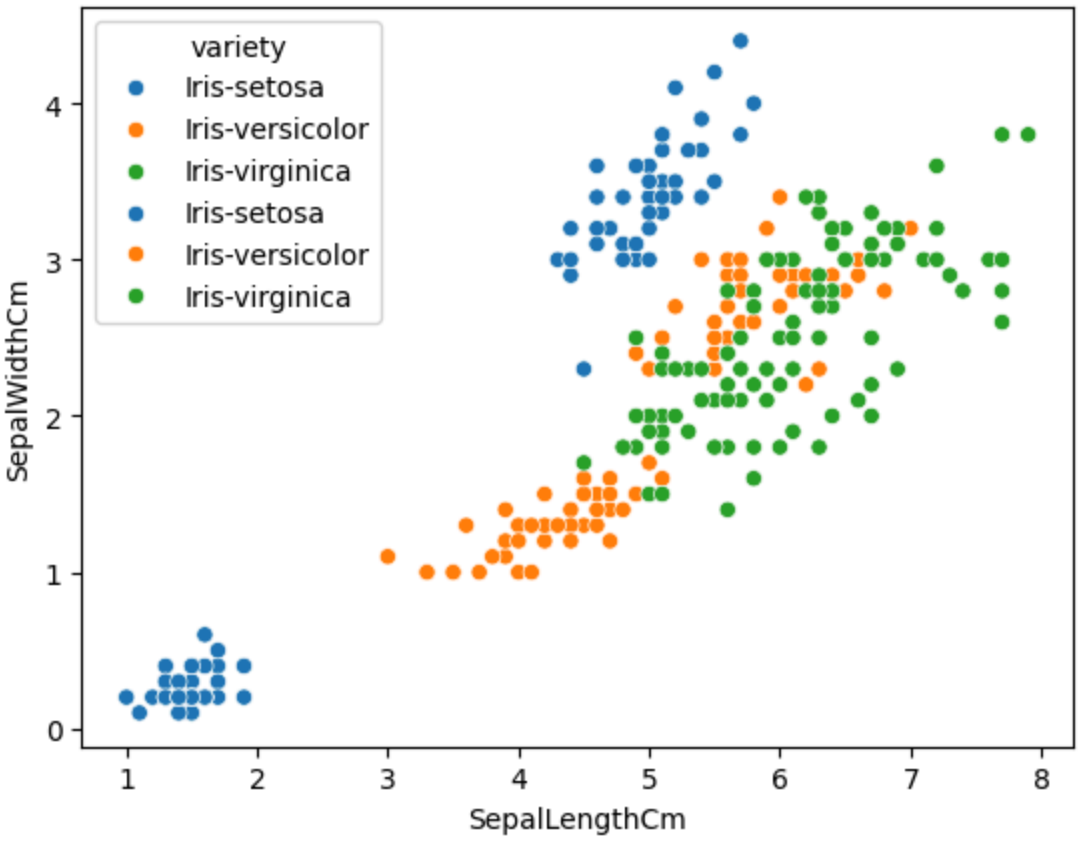
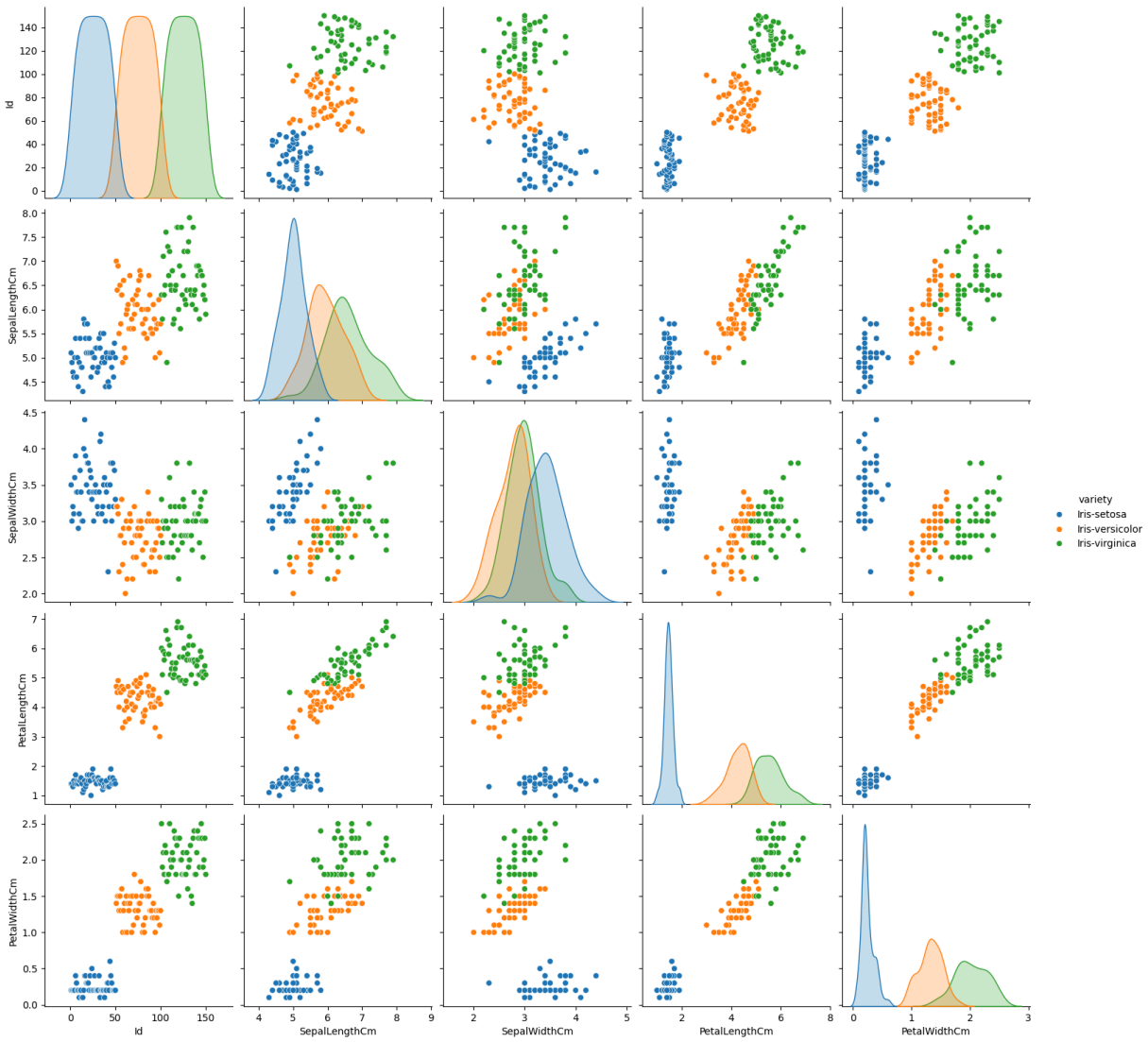
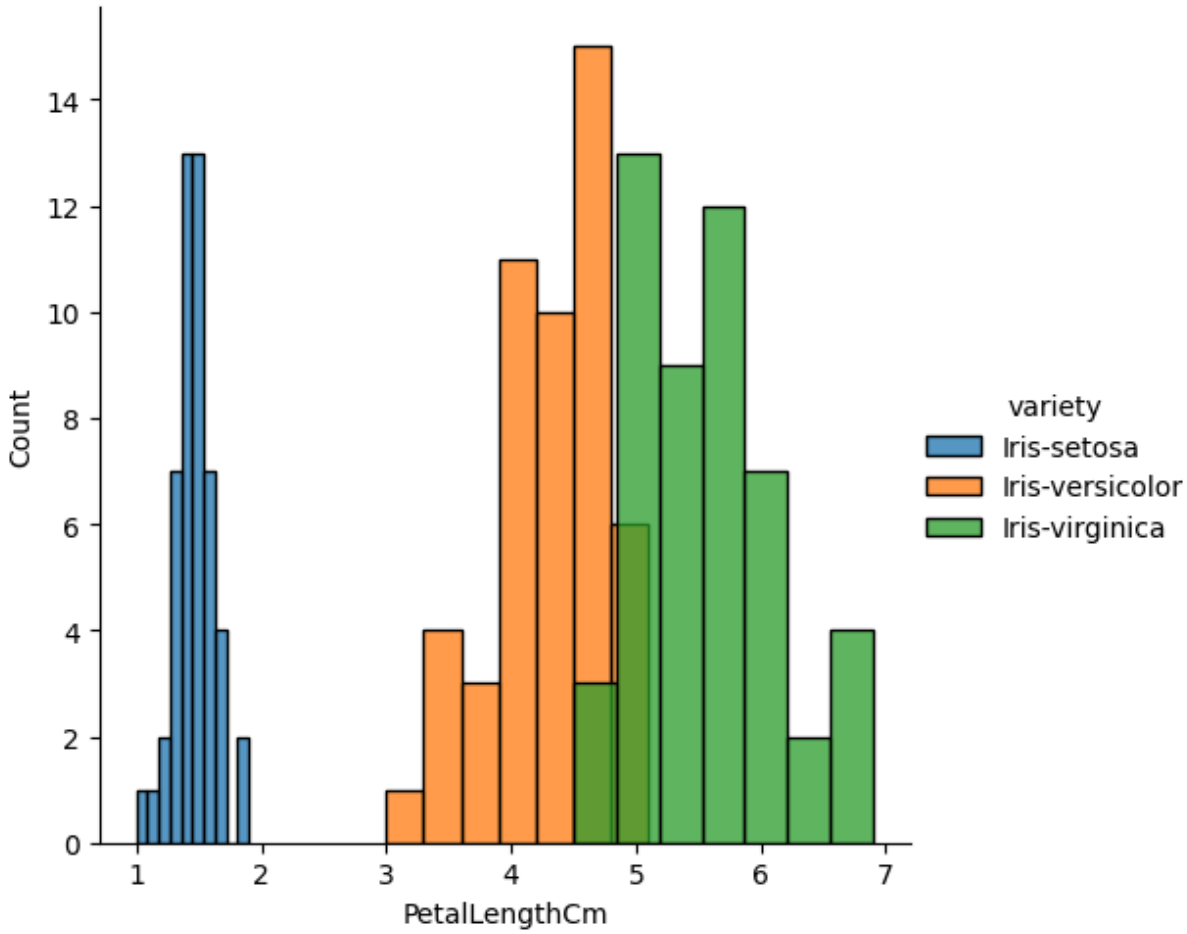
In [1]:
```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
%matplotlib inline
data=pd.read_csv('/content/Iris_Dataset.csv')
data
data.info()
data.describe()
data.value_counts('variety')
sns.countplot(x='variety',data=data,)
plt.show()
dummies=pd.get_dummies(data.variety)
FinalDataset=pd.concat([pd.get_dummies(data.variety),data.iloc[:,[0,1,2,3]]],axis=1)
FinalDataset.head()
sns.scatterplot(x='SepalLengthCm',y='SepalWidthCm',hue='variety',data=data,)
sns.scatterplot(x='PetalLengthCm',y='PetalWidthCm',hue='variety',data=data,)
sns.pairplot(data,hue='variety',height=3);
plt.show()
sns.FacetGrid(data,hue='variety',height=5).map(sns.histplot,'PetalLengthCm').add_legend();
plt.show();
sns.FacetGrid(data,hue='variety',height=5).map(sns.histplot,'PetalWidthCm').add_legend();
plt.show();
sns.FacetGrid(data,hue='variety',height=5).map(sns.histplot,'SepalLengthCm').add_legend();
plt.show();
sns.FacetGrid(data,hue='variety',height=5).map(sns.histplot,'SepalWidthCm').add_legend();
plt.show();
```

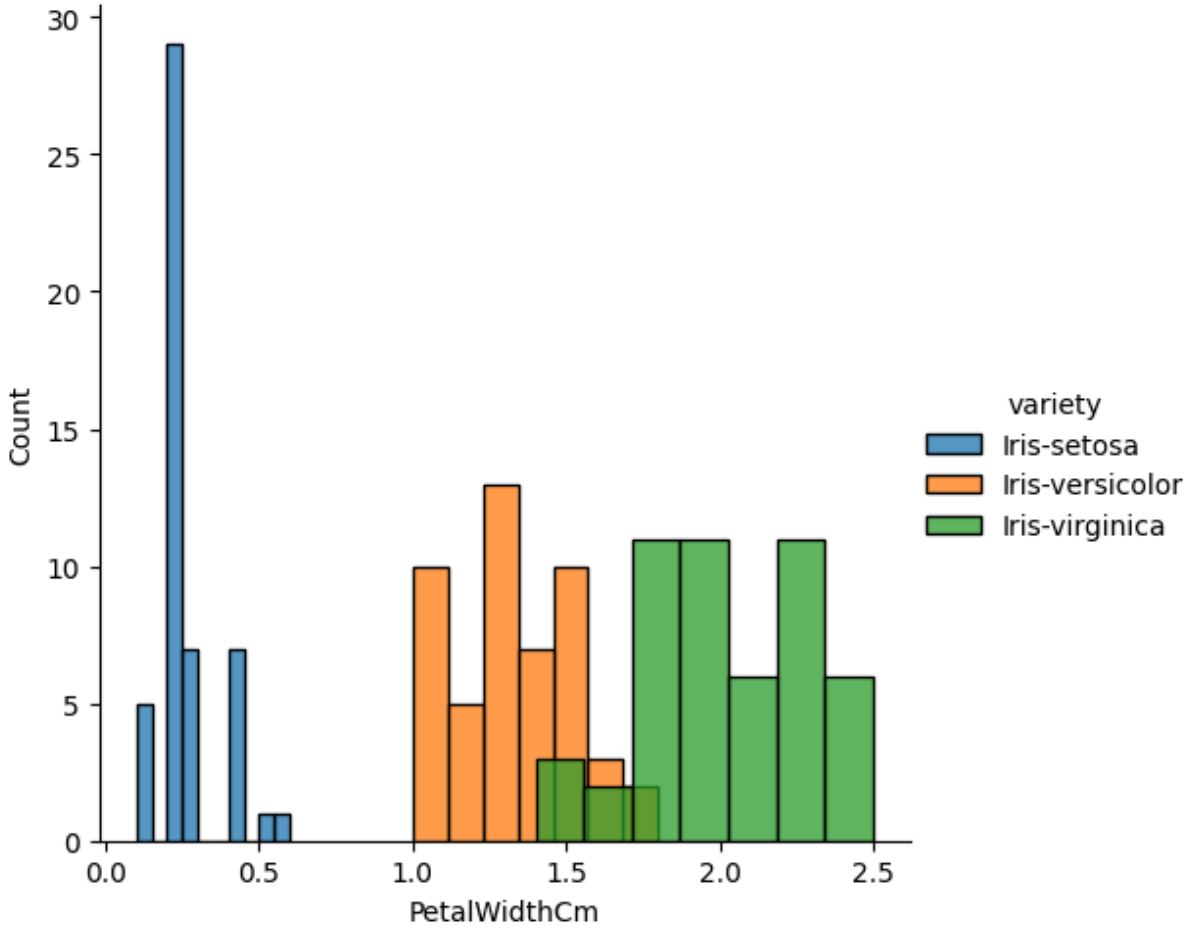
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 6 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Id             150 non-null    int64
 1   SepalLengthCm  150 non-null    float64
 2   SepalWidthCm   150 non-null    float64
 3   PetalLengthCm  150 non-null    float64
 4   PetalWidthCm   150 non-null    float64
 5   variety        150 non-null    object
dtypes: float64(4), int64(1), object(1)
memory usage: 7.2+ KB
```

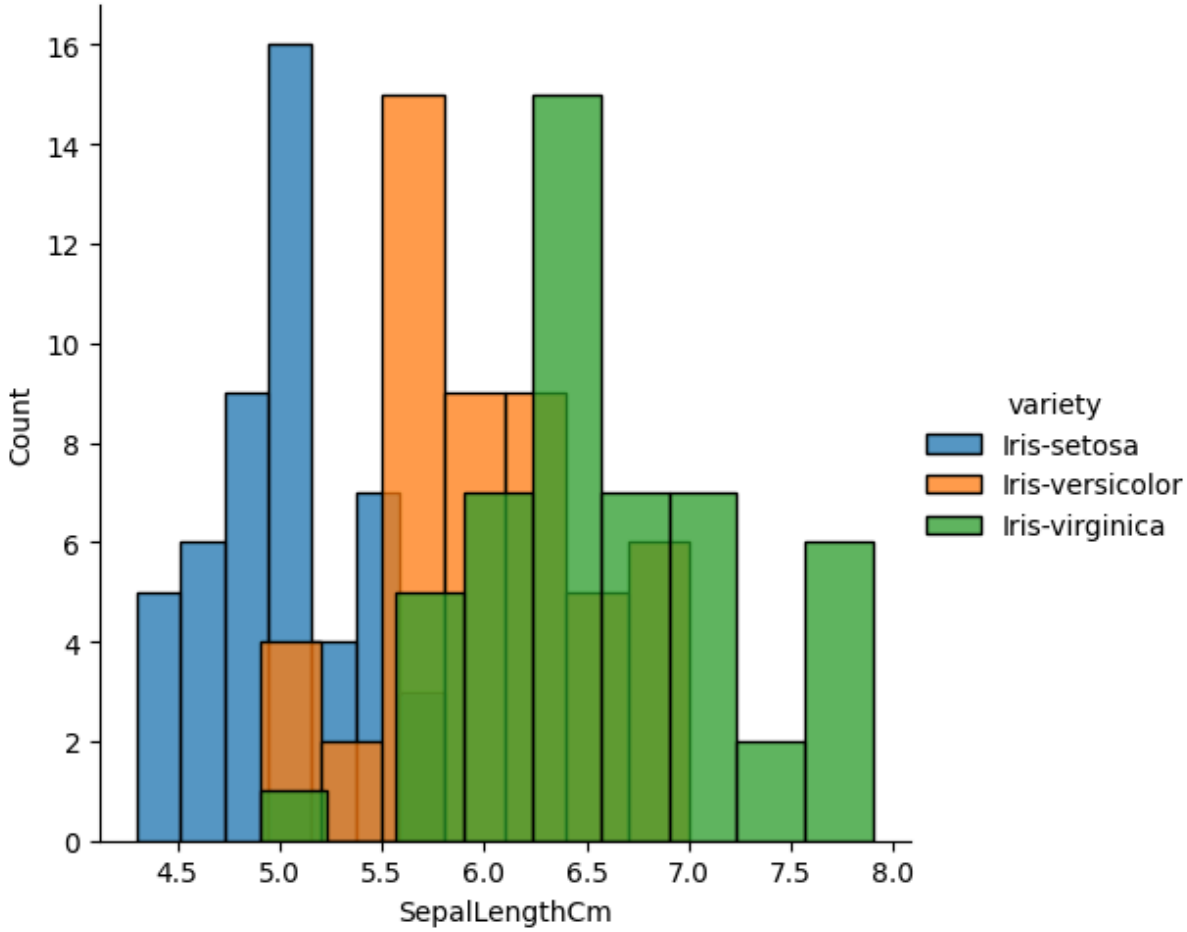

50
40
30
20
10
0
count
Iris-setosa
Iris-versicolor
Iris-virginica
variety

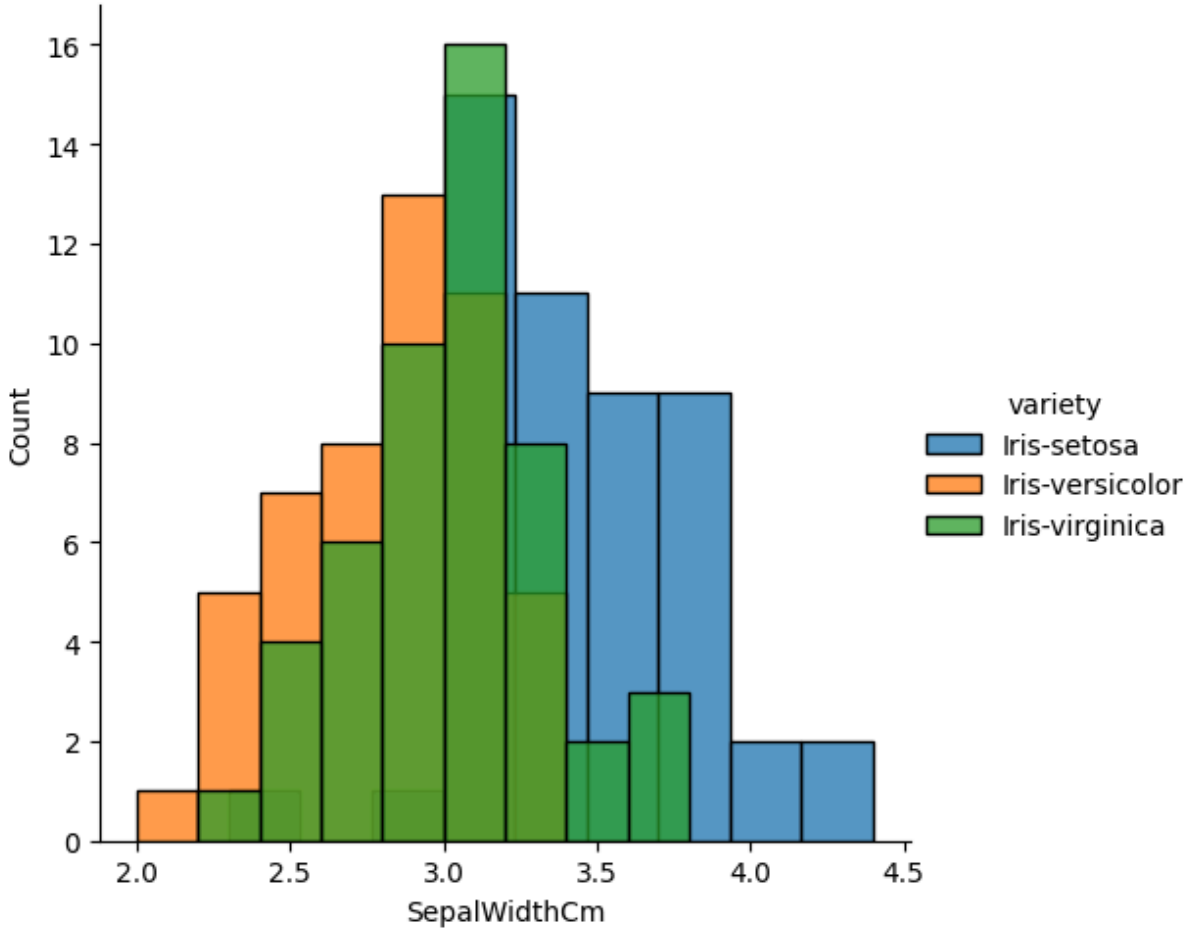

variety
Iris-setosa
Iris-versicolor
Iris-virginica
Iris-setosa
Iris-versicolor
Iris-virginica
SepalWidthCm
SepalLengthCm
0
1
2
3
4
1
2
3
4
5
6
7
8


Id
SepalLengthCm
SepalWidthCm
PetalLengthCm
PetalWidthCm
variety
Iris-setosa
Iris-versicolor
Iris-virginica


Count
PetalLengthCm
variety
Iris-setosa
Iris-versicolor
Iris-virginica


30
25
20
15
10
5
0
Count
0.0
0.5
1.0
1.5
2.0
2.5
PetalWidthCm
variety
Iris-setosa
Iris-versicolor
Iris-virginica


Count
SepalLengthCm
variety
Iris-setosa
Iris-versicolor
Iris-virginica


Count
SepalWidthCm
variety
Iris-setosa
Iris-versicolor
Iris-virginica